

A. Michael West Jr., PhD – Post-doctoral Fellow, Johns Hopkins University, Mechanical Engineering

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RESEARCH INTERESTS

My research, rooted in control theory, advances robotic technologies such as rehabilitation robotics, collaborative robots, prosthetics, and dexterous robots by investigating human motor control, learning, and perception. Combining expertise in controls, robotics, biomechanics, and neuroscience, I develop simple, descriptive motor control models that offer immediate and easily interpretable insight. Introducing novel analysis methods for human motor control and leveraging established approaches on new experimental paradigms, my work consistently uncovers fresh insights on human motor control, learning, and perception.

EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, Massachusetts

Ph.D., Mechanical Engineering, May 2024, GPA: 5.0/5.0

- Dissertation: [All Models are Wrong. Simple Models Provide Insight: A Study of Human Manipulation](#)
- Advisor: Neville Hogan
- Committee: Haruhiko Harry Asada, Alberto Rodriguez, and Madhusudhan Venkadesan

M.S., Mechanical Engineering, May 2020, GPA: 4.5/5.0

- Dissertation: Towards a non-invasive measurement of human motion, force, and impedance during a complex physical-interaction task: wire-harnessing.
- Advisor: Neville Hogan

Yale University, New Haven, Connecticut

B.S., Mechanical Engineering, May 2018, GPA: 3.5/4.0, Major GPA: 3.6/4.0

HONORS & AWARDS

Edward A. Bouchet Graduate Honor Society, Johns Hopkins University	2026
Johns Hopkins University Provost's Postdoctoral Fellowship Program (PPF) (\$97,475)	2025
Johns Hopkins University Provost's Postdoctoral Fellowship Program (PPF) (\$100,880)	2024
Office of Graduate Education Service Award, MIT Recognizing Individuals Supporting Equity Awards	2024
Shirley Anne Jackson Award, MIT Ebony Affair	2024
• Presented to the black graduate student who exemplifies the spirit of trailblazing leadership, academic excellence, and commitment to breaking barriers in their field of study.	
Inclusive Teaching Certificate, MIT Teaching + Learning Lab	2023
12 th Place (out of 300+ Submissions) Best Poster Competition, Rehab Week	2023
Accenture Fellowship (\$98,349) , MIT	2023
MechE Research Exhibition (MERE), Honorable Mention Best Overall Performance (\$125) , MIT	2023
BAMIT Research SLAM, 2 nd Place (\$500) , Black Alumni of MIT (BAMIT)	2023
Takeda Fellowship (\$94,789) , MIT	2022
Ben Gold Fellowship (\$90,932.76) , MIT	2021
Ford Foundation Fellowship (\$81,000) , Ford Foundation	2020
Teaching Practice Certificate, MIT Teaching + Learning Lab	2020
Black Graduate Student Association (BGSA) Excellence in Research Award, MIT Ebony Affair	2020
Subject Design Certificate, MIT Teaching + Learning Lab	2020
GEM University Associate Fellowship, MIT	2018 & 2020
Ford Foundation Fellowship Honorable Mention, Ford Foundation	2018
UCEM SLOAN Scholarship (\$40,000) , MIT	2018
Office of Graduate Education (OGE) Diversity Fellowship (\$80,872) , MIT	2018
Student Design Showcase, 3 rd Place, Design of Medical Devices Conference	2017

RESEARCH EXPERIENCE

Johns Hopkins University, Haptics and Medical Robotics (HAMR) Laboratory

Advisor: Jeremy Brown

Provost's Postdoctoral Research Fellow

2024 – Present

- Developing and evaluating rehabilitation strategies to mitigate sensorimotor impairments in wrist and hand function.
- Investigating the development of surgeon visual-haptic acuity during robot-assisted minimally invasive surgery, with implications for training and robotic system design.
- Examining how the nervous system coordinates, or dissociates, motor control and motor prediction during sensorimotor learning.
- Mentoring and advising PhD, master's, undergraduate, and high school students on research methodologies, experimental design, and career development.

Massachusetts Institute of Technology, Newman Laboratory for Biomechanics and Human Rehabilitation

Advisor: Neville Hogan

Graduate Student

2018 – 2024

My research uses analysis of kinematic data recorded during human behavioral studies to unveil latent aspects of human motor control, learning, and perception for applications in rehabilitation and robotics. An emphasis is placed on the development of simple, descriptive models.

- Explored how kinematic analysis of the human hand can be used to estimate the complexity of a dexterous task.
- Collected & analyzed data on human hand manipulation of complex objects & tools.
- Investigated how humans estimate hidden mechanical properties of another's limbs simply by observing their motions. Developed a model of human motor perception.
- Conducted a human subject experiment to analyze how humans control force and motion during human-robot physical interaction.

Yale University, Biomechanics & Control Lab

Advisor: Madhusudhan Venkadesan

Undergraduate Research Assistant

Fall 2017 – Spring 2018

- Designed & conducted an experiment to validate a geometric theory that estimated limb impedance using joint surface features.

Yale University, Medical Device Design & Innovation

Advisor: Joseph Zinter

Design Engineering Student

Fall 2016 – Fall 2017

- Designed & built a 125-part saw guide to improve sternotomies & decrease complications stemming from sternal non-union.

Yale University, Social Robotics Laboratory

Advisor: Brian Scassellati

Undergraduate Research Assistant

Summer 2016

- Designed & built the interior & exterior components of a social robot's articulating face to improve infant-robot interaction.

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **West, A. M.**, Miller, A., Haith, A., & Brown, J. (Under Review). Dissociation of Prediction and Control During Implicit Adaptation. *Journal of Neurophysiology*.
2. Pimpalkar, A., **West, A. M.**, Rai, D., Xu, J., & Brown, J. (Under Review). Evaluating Calibrated Multimodal Feedback in Augmenting Motor Rehabilitation. *Journal of NeuroEngineering and Rehabilitation (JNER)*.
3. Huang, T., Huber, M., Brown, J. & **West, A. M.** (Under Review). [Perceiving latent dynamics: Innate and coachable visual estimation of limb damping](#). *Proceedings of the National Academy of Sciences (PNAS)*.
4. Tessari, F., **West, A. M.** & Hogan, N. (2025). [Explaining human motor coordination via the synergy expansion hypothesis](#). *The Proceedings of the National Academy of Sciences (PNAS)*. [[Video Summary](#)]
5. **West, A. M.** & Hogan, N. (2025). [Kinematic hand synergies differ between reach-and-grasp and functional object manipulation](#). *Journal of Neurophysiology*.
6. **West, A. M.**, Tessari, F., Wang, M., & Hogan, N. (2025) [The Study of Dexterous Hand Manipulation: A Synergy-Based Complexity Index](#). *IEEE Transactions on Medical Robotics and Bionics*. [[Video Summary](#)]
7. Lachner, J., Tessari, F., **West, A. M.**, Nah, M.C., & Hogan, N. (2024). Divide et Impera: Learning impedance families for peg-in-hole assembly. <https://arxiv.org/html/2410.01054v1>.
8. **West, A. M.**, Huber, M.E., & Hogan, N. (2022). [Role of Path Information in Visual Perception of Joint Stiffness](#). *PLOS Computational Biology*. [[Video Summary](#)]
9. **West, A. M.**, Hermus, J., Huber, M., Maurice, P., Sternad, D., & Hogan, N. (2022). [Dynamic Primitives Limit Human Force Regulation during Motion](#). *IEEE Robotics and Automation Letters (RA-L)*. [[Video Summary](#)]

PEER-REVIEWED CONFERENCE PUBLICATIONS

1. **West Jr., A. M.**, Kasmican, B., Pimpalkar, A., Xu, J., & Brown, J. (2026). Cross-Modal Matching Is Not Bidirectional: Implications for Human Multisensory Feedback. *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*.
2. **West, A. M.**, Pimpalkar, A., Xu, J., & Brown, J. (2025). [SenseMatch: Smartphone-Based Cross-Modal Matching for Accessible Perceptual Assessments](#). *IEEE EMBS Conference on Neural Engineering (NER)*.
3. Tessari, F., **West, A. M.** & Hogan, N. (2025). [A Geometric Approach for the Comparison of Kinematic Synergy Postures](#). *IEEE International Conference on Rehabilitation Robotics (ICORR)*.
4. Pimpalkar, A., **West, A. M.**, Xu, J., & Brown, J. (2025). [Optimizing Cross-Modal Matching for Multimodal Motor Rehabilitation](#). *IEEE International Conference on Rehabilitation Robotics (ICORR)*.
 - a. **Best Student Paper Award Finalist**
5. **West, A. M.**, Tessari, F., & Hogan, N. (2023). [The Study of Complex Manipulation via Kinematic Hand Synergies: The Effects of Data Pre-Processing](#). *IEEE International Conference on Rehabilitation Robotics (ICORR)*. [[Video Summary](#)]

SHORT PEER-REVIEWED CONFERENCE ARTICLES & ABSTRACTS

1. **West Jr., A. M.**, Kasmican, B., Pimpalkar, A., Xu, J., & Brown, J. (2026) Cross-modal matching is not bidirectionally invertible: Implications for multisensory representation. Poster presented at the 2021 *Society for the Neural Control of Movement Conference (NCM)*, Kobe, Japan.
2. Tessari, F., Happold, J., **West Jr., A. M.**, Capsi-Morales, P., Lachner, J., Hogan, N., & Piazza, C. (2026) The role of kinematic and muscular synergies in dexterous, contact-rich manipulation. Poster presented at the 2021 *Society for the Neural Control of Movement Conference (NCM)*, Kobe, Japan.
3. Huang, T., Brown, J. & **West, A. M.** (Under Review). Perceiving latent dynamics: Innate and coachable visual estimation of limb damping. Poster presented at the 2021 *Society for the Neural Control of Movement Conference (NCM)*, Kobe, Japan.
4. **West, A. M.**, Pimpalkar, A., Xu, J., & Brown, J. (2025). SenseMatch: Smartphone-Based Cross-Modal Matching for Accessible Perceptual Assessments. Poster presented at the *IEEE EMBS Conference on Neural Engineering (NER)*.

5. Berthet, M., **West Jr, A. M.**, Branscheidt, M., & Awai, C. E. (2025). [VAF-CI: Quantifying low-dimensional global movement complexity during a novel lower limb motor learning task](#) *Gait & Posture*
6. **West, A. M.**, Huber, M.E., & Hogan, N. (2025). Role of Path Information in Visual Perception of Joint Stiffness. PowerPoint presented at the *Summer School on Neurorehabilitation (SSNR)*, Vigo, Spain.
7. **West, A. M.**, Tessari, F., Wang, M., & Hogan, N. (2025) The Study of Dexterous Hand Manipulation: A Synergy-Based Complexity Index. Poster presented at the *Summer School on Neurorehabilitation (SSNR)*, Vigo, Spain.
8. **West, A. M.**, Tessari, F., Wang, M., & Hogan, N. (2024) The Study of Dexterous Hand Manipulation: A Synergy-Based Complexity Index. PowerPoint presented at *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, Heidelberg, Germany.
9. **West, A. M.**, Tessari, F. & Hogan, N. (2023). The Study of Complex Manipulation via Kinematic Hand Synergies: The Effects of Data Pre-Processing. PowerPoint & poster presented at *Rehab Week*, Singapore. **12th place (out of 300+ submissions) best poster award.**
10. **West, A. M.**, & Hogan, N. (2023). Kinematic Hand Synergies Differ during Tool-Use and Object Manipulation. Poster presented at the 2023 *Society for Neural Control of Movement (NCM) Conference*, Victoria, BC, Canada.
11. **West, A. M.**, Hermus, J., Huber, M., Maurice, P., Sternad, D., & Hogan, N. (2022). Dynamic Primitives Limit Human Force Regulation during Motion. PowerPoint & poster presented at the 2022 *International Conference of Robotics and Automation (ICRA)*, Philadelphia, PA, USA.
12. **West, A. M.**, Hermus, J., Huber, M., Maurice, P., Sternad, D., & Hogan, N. (2022). Dynamic Primitives Limit Human Force Regulation during Motion. Poster presented at the 2022 *MIT Sports Summit*, Cambridge, MA, USA.
13. **West, A. M.**, Huber, M.E., Hermus, J., Maurice, P., Sternad, D., & Hogan, N. (2021). Humans Do Not Directly Control Force During Physical Interaction. Poster presented at the 2021 *Society for the Neural Control of Movement Conference (NCM)*, Virtual.
14. **West, A. M.**, Huber, M.E., & Hogan, N. (2020). Role of Path Information in Visual Perception of Joint Stiffness. PowerPoint presented at the 2020 *Neuromatch Conference*, Virtual.
15. **West, A. M.**, Huber, M.E., Hermus, J. & Hogan, N. (2020). Humans Do Not Directly Control Force During Physical Interaction. PowerPoint presented at the 2020 *Conference of Ford Fellows*, Virtual.
16. **West, A. M.**, & Hogan, N. (2020). An Analysis of Kinematic Hand Synergies During Wire-Harness Installation. PowerPoint presented at the 2020 *International Conference of Robotics and Automation (ICRA)*, Virtual.
17. Huber, M.E., **West, A. M.**, Folinus, C., & Hogan, N. (2019). Visual Perception of Joint Stiffness from Multi-Joint Limb Motion. Poster presented at the 2019 *Annual Conference of the Society for Neuroscience (SfN)*, Chicago, IL, USA.
18. **West, A. M.**, Mandl, H., McCann, C., & Gunawardena, N. (2017). A Novel Sternotomy Saw Guide Incorporating Integrated Rigid Fixation. Poster presented at the 2017 *Design of Medical Devices Conference*, Minneapolis, MN, USA.

INVITED TALKS

1. **“Latent Structures in Human Motor Control and Perception: Implications for Robotic Coaches”** National Symposium on Effective & Ethical AI (NSEA) Human-AI Collaboration Empowerment, Agency, and Belonging, Morgan State University, July 26th, 2026
2. **“From Athlete to Scientist: Understanding and Improving Human Movement Through Robotics”** Massachusetts Institute of Technology Summer Research Program (MSRP), July 23rd, 2026
3. **“Michael West’s Journey to Dr. Michael West, PhD”** Promoting Research Opportunities in Engineering Labs (PROPEL), Johns Hopkins University, June 30th, 2026
4. **“Latent Structures in Human Motor Control and Perception: Implications for Skill Acquisition”** Malone Center Research Luncheon, Johns Hopkins University, May 14th, 2026

5. **"Fundamentals Unifying Neuromotor Control and Robotics"** Georgia Tech Safe, Scalable, and Sustainable Human-AI-Robot Teaming: Interaction, Synergy, and Augmentation, May 5th, 2026
6. **["How We See, How We Feel: Exploring Human Perception"](#)** Georgia Tech's Online Master of Science in Computer Science (OMSCS) program, September 18th, 2025
7. **"Undergraduate Research as a Launchpad: Insights from a Post-doctoral Fellow"** Advisor Open House, Johns Hopkins University, August 13th, 2025
8. **"Michael West's Journey to Dr. Michael West, PhD"** Promoting Research Opportunities in Engineering Labs (PROPEL), Johns Hopkins University, June 25th, 2025
9. **"Navigating the Faculty Interview Process: Advice for Candidates"** Postdoctoral Research Association, Johns Hopkins University, February 18th, 2025
10. **"All Models are Wrong, Simple Models are Insightful: A Study of Human Manipulation"** Department of Mechanical Engineering, Johns Hopkins University, September 19th, 2024
11. **"The Newman Laboratory for Biomechanics, Robotics and Human Rehabilitation: A journey through motor coordination, manipulation and physical interaction"** Munich Institute of Robotics and Machine Intelligence (MIRMI), Technical University of Munich, June 17th, 2024
12. **"All Models are Wrong, Simple Models are Insightful: A Study of Human Manipulation"** Department of Mechanical Engineering, Massachusetts Institute of Technology, April 11th, 2024
13. **"All Models are Wrong, Simple Models are Insightful: A Study of Human Manipulation"** Department of Robotics, University of Michigan, March 28th, 2024
14. **"All Models are Wrong, Simple Models are Insightful: A Study of Human Manipulation"** Department of Electrical Engineering and Computer Sciences, University of California Berkeley, February 27th, 2024
15. **"All Models are Wrong, Simple Models are Insightful: A Study of Human Manipulation"** George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology, February 14th, 2024
16. **"Exploring Human Performance: The Importance of Studying our Limitations"** Collaborative Haptics and Robotics in Medicine Lab (Allison Okamura), Stanford University, May 10th, 2023
17. **["Exploring Human Performance: The Importance of Studying our Limitations"](#)** Neuromuscular Biomechanics Lab (Scott Delp), Stanford University, May 10th, 2023
18. **"Understanding Human Neuromotor Control: All Models are Wrong, Simple Models are Useful"** Neurobionics Engineering Club, Camplus College, May 4th, 2023
19. **["Mechanical Impedance: A Necessity in Both Human Motor Action and Human Motor Perception"](#)** Control Conclave, Indian Institute of Technology Delhi, January 6th, 2023
20. **["Mechanical Impedance: A Key Component in Both Human Motor Action and Motor Perception"](#)** Future Leaders in Mechanical and Aerospace Engineering: Celebrating Diversity and Innovation, March 10th, 2021

TEACHING EXPERIENCE

Johns Hopkins University, 530.343 Design and Analysis of Dynamical Systems

Co-Instructor with Prof. Louis Whitcomb

Spring 2025

- Independently developed and delivered a series of structured lectures as part of a required undergraduate course, creating concept-driven lessons that complemented the broader instructional plan.
- Collaborated with the lead instructor to shape the course's direction and learning outcomes; co-developed assessments including quizzes, problem sets, and exams aligned with course objectives.
- Taught core topics such as modeling and analysis of damped and undamped, forced and free vibrations in single and multi-degree-of-freedom linear dynamical systems, as well as introductory concepts in system stability and control.

Massachusetts Institute of Technology, Teaching + Learning Lab Inclusive Teaching Track

Fall 2023

- Investigated research on “belonging” in academic settings & developed actionable strategies to cultivate a sense of belonging in teaching contexts.
- Devised discipline specific teaching practices that consistently engage in critical reflection to assess & enhance inclusivity within my classroom.

Massachusetts Institute of Technology, 2.183J/4 Biomechanics & Neural Control of Movement

Teaching Assistant to Professor Neville Hogan

Spring 2021

- Assisted students in understanding homework & lecture material via weekly office hours.
- Helped organize the course & develop lessons based on the courses teaching topics.
- Orchestrated a course designed for upper-level undergraduate students and graduate students. It was comprised of 7 undergraduate & 15 graduate students.
- Presented students with a quantitative description of how biomechanical & neural factors interact in human sensory-motor behavior. Topics included a review of relevant neural, muscular & skeletal physiology, neural feedback & "equilibrium-point" theories, co-contraction strategies, impedance control, kinematic redundancy, optimization, intermittency, contact tasks & tool-use.

Massachusetts Institute of Technology, 2.151 Advanced System Dynamics & Control

Teaching Assistant to Professor Neville Hogan

Fall 2020

- Independently led students in weekly problem-solving based lectures.
- Helped organize the course & develop lessons based on the courses teaching topics.
- Support a course designed for graduate students. It was comprised of 30 graduate students.
- Educated topics including analytical descriptions of state-determined dynamic physical systems, time & frequency domain representations, system characteristics, & modification of system characteristics using feedback.

Massachusetts Institute of Technology, Teaching + Learning Lab

Teaching Practice Certificate Program

Summer 2020

- Discovered several evidence-based teaching strategies that help students stay cognitively active & enable them to practice the material & immediately test their knowledge.
- Practiced giving a lecture that incorporated active learning techniques on a topic involving computer programming.

Massachusetts Institute of Technology, Teaching + Learning Lab

Subject Design Certificate Program

Summer 2020

- Practiced defining a lecture’s scope, goals, & objectives, & how to effectively assess student progress & give useful feedback.
- Developed an ability to cultivate a classroom that values a diverse set of thoughts, perspectives, & experiences by acknowledging my own experiences & biases.

SUPERVISION OF GRADUATE MENTEES

Vibha Kamath, M.S., Mechanical Engineering, JHU

Spring 2026 – Present

Aaliyah Mazzeo, Ph.D., Biomedical Engineering, JHU

Summer 2025 – Present

Berk Kasmican, M.S., Mechanical Engineering, JHU

Summer 2025 – Present

Next Position: PhD Candidate, Vanderbilt University

Publications:

- West Jr., A. M., **Kasmican, B.**, Pimpalkar, A., Xu, J., & Brown, J. (2026). Cross-Modal Matching Is Not Bidirectional: Implications for Human Multisensory Feedback. *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*.

Anway Pimpalkar, M.S., Biomedical Engineering, JHU

Fall 2024 – Summer 2025

Next Position: PhD Candidate, Harvard University

Publications:

- **Pimpalkar, A.**, West, A. M., Rai, D., Xu, J., & Brown, J. (Under Review). Evaluating Calibrated Multimodal Feedback in Augmenting Motor Rehabilitation. *Journal of NeuroEngineering and Rehabilitation (JNER)*.
- West Jr., A. M., Kasmican, B., **Pimpalkar, A.**, Xu, J., & Brown, J. (2026). Cross-Modal Matching Is Not Bidirectional: Implications for Human Multisensory Feedback. *IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*.
- West, A. M., **Pimpalkar, A.**, Xu, J., & Brown, J. (2025). SenseMatch: Smartphone-Based Cross-Modal Matching for Accessible Perceptual Assessments. *IEEE EMBS Conference on Neural Engineering (NER)*.
- **Pimpalkar, A.**, West, A. M., Xu, J., & Brown, J. (2025). Optimizing Cross-Modal Matching for Multimodal Motor Rehabilitation. *IEEE International Conference on Rehabilitation Robotics (ICORR)*.
- **Best Student Paper Award Finalist**

Margaret Wang, M.Eng., Electrical Engineering & Computer Science, MIT

Fall 2022 – Spring 2023

Next Position: Roboticist, Boston Dynamics AI Institute

Publications:

- **Wang, M. X.** (2023). [Non-invasive vision-based measurement of hand kinematics and interaction](#) (M. Eng, MIT).
- West, A. M., Tessari, F., **Wang, M.**, & Hogan, N. (2025) The Study of Dexterous Hand Manipulation: A Synergy-Based Complexity Index. *IEEE Transactions on Medical Robotics and Bionics*.

SUPERVISION OF UNDERGRADUATE MENTEES

Maximilian Moody, Mechanical Engineering, UMBC

Summer 2026

Lydia Wosen, Mechanical Engineering, Howard University

Summer 2026

Taliyah Huang, Computer Science and Biomedical Engineering, JHU

Spring 2025 – Present

Publications:

- **Huang, T.**, Huber, M., Brown, J. & West, A. M. (Under Review). Perceiving latent dynamics: Innate and coachable visual estimation of limb damping. *PLOS Computational Biology*.

Caly Ferguson, Mechanical Engineering, UMBC

Summer 2025

Next Position: PhD Candidate, Johns Hopkins University

Jason Childs, Electrical Engineering, Howard University

Summer 2025

Next Position: Summer Undergraduate Research Fellow (SURF), Caltech

Nicole Attram, Mechanical Engineering, UMBC

Summer 2023 – Fall 2024

Next Position: PhD Candidate, Northwestern University

Shreya Gupta, Mechanical Engineering, MIT

Summer 2020 – Fall 2020

SUPERVISION OF HIGH SCHOOL MENTEES

Naomi M., Garrison Forest School

Spring 2025

WORK EXPERIENCE

Medtronic, North Haven

Robotic Design Intern

Summer 2018

- Designed & built the first prototype of an articulating endoscope to interface with a surgical robot, using Creo Parametric 3.0.
- Implemented the endoscopes controller model using Simuscape 2017.

Yale University, Yale Student Technology Collaborative (STC)

Computer Lab Technician

2017 – 2018

- Repaired hardware & software issues of on-campus lab computers.

**Yale University, Center for Engineering Innovation & Design (CEID)
Design Aide**

2017 – 2018

- Provided theoretical & technical expertise to students seeking help on design projects in the CEID.

COMMUNITY SERVICE & LEADERSHIP

Johns Hopkins University, Mechanical Engineering Department Belonging Committee

Post-Doctoral Representative

Spring 2025 – Present

- Contributes to initiatives that support student well-being, promotes inclusive recruitment, and informs departmental strategies to foster an equitable academic environment.

Massachusetts Institute of Technology, New Vassar Dormitory

Graduate Resident Advisor

Fall 2020 – Spring 2024

- Fostered a supportive, safe, & positive living environment for 38 students through events built upon shared community standards.
- Completed numerous training courses on student support, facilitating group dynamics, fostering interpersonal relationships, and mandatory reporting as a responsible employee.

Massachusetts Institute of Technology, Black Graduate Student Association (BGSA)

Treasurer

Spring 2021 – Spring 2022

- Allocated & managed funds (~\$21,000) to hold community service & social events for MIT's Black graduate students.

Massachusetts Institute of Technology, MIT Summer Research Program (MSRP)

Pod Leader

Summer 2019 & Summer 2021

- Organized meetings & events to foster a positive climate for undergraduate minority students conducting research at MIT.
- **16 of the 19 students I mentored have enrolled in top STEM graduate programs including MIT, Stanford, UC Berkeley, Georgia Tech, UCLA, & Purdue.**

Massachusetts Institute of Technology, Academy of Courageous Minority Engineers (ACME)

Treasurer

Spring 2019 – Spring 2022

- Allocated & managed funds (~\$8,000) for a group of minority engineers who met weekly to discuss career goals & hold each other accountable in achieving those goals.

Massachusetts Institute of Technology, Men's Rugby Football Club

President & Captain

Spring 2019 – Spring 2024

- Oversaw team operations, organized competitive events, & fostered a cohesive team environment, resulting in elevated team performance & strengthened camaraderie.
- Initiated & led comprehensive post-COVID revitalization efforts to reignite team participation & boost morale, resulting in a significant increase in club membership & active participation.

Yale University Admissions Office

STEM Outreach Ambassador

Spring 2017 – Spring 2018

- Mentored high school students, who had expressed interest in pursuing STEM fields, in making an informed college decision.

National Society of Black Engineers (NSBE) Yale University Chapter

Vice President

Spring 2016 – Spring 2017

- **Founded a tutoring service at New Haven Academy High School.**
- Facilitated chapter meetings & increased comradery through encouraging Black students to continue pursuing STEM fields.
- Assisted in managing organization finances in efforts to sponsor members to attend the national conferences.

Yale University Football

Defensive End & Linebacker

Fall 2014 – Summer 2016

- Devoted over 30 hours per week to film study, practice sessions, rigorous strength & conditioning, & encouraging teammates to foster team morale.

PROFESSIONAL SERVICE

Journal Paper Reviews

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)
- IEEE Transactions on Biomedical Engineering (TBME)
- PLOS Computational Biology
- Scientific Reports
- Qeios
- npj Soft Matter

Conference Paper Reviews

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference for Rehabilitation Robotics (ICORR)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- Robotics: Science and Systems (RSS)

Workshops Organized

- “Beyond Task Performance: Designing Robots for Human Motor Learning and Rehabilitation” 2026 IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)
- “Human-Inspired Principles for Robotic Dexterous Manipulation: What are we Missing?” 2026 IEEE International Conference on Robotics and Automation (ICRA)
- “Is open-source hardware the road forward for Upper Extremity Rehabilitation and Neuroscience?”, *2025 IEEE International Consortium for Rehabilitation Robotics (ICORR)*

PROFESSIONAL DEVELOPMENT

Yale Bouchet Conference on Graduate Education, New Haven, CT	April 2026
National Society of Black Engineers (NSBE) Annual Convention, Baltimore, MD	March 2026
Academics with Diversity Education and Mentorship in Engineering, University of Akron	July 2025
The Summer School on Neurorehabilitation (SSNR), Vigo, Spain	June 2025
The Inaugural Raynor Cerebellum Project Postdoc Summit, Scottsdale, AZ	November 2024
The Institute on Teaching and Mentoring, New Orleans, LA	October 2024
BE GREAT Retreat (Hosted by Dr. Brooke Coley), Atlanta, GA	March 2024
National Society of Black Engineers (NSBE) Annual Convention, Atlanta, GA	March 2024
The Institute on Teaching and Mentoring, Tampa, FL	October 2023
Trail Blazers in Engineering, Purdue University	July 2023
Postdoctoral Recruitment Initiative in Sciences and Medicine (PRISM), Stanford University	May 2023
Focus Fellows Program, Georgia Institute of Technology	January 2023
The Institute on Teaching and Mentoring, Atlanta, GA	October 2022

NextProf Nexus Future Faculty Workshop, University of California, Berkeley	September 2022
The Institute on Teaching and Mentoring, Atlanta, GA	October 2019
National Society of Black Engineers (NSBE) Annual Convention, Pittsburgh, PA	March 2018
National Society of Black Engineers (NSBE) Annual Convention, Kansas City, MO	March 2017
National Society of Black Engineers (NSBE) Annual Convention, Boston, MA	March 2016

IN THE NEWS

10 Johns Hopkins PhDs and postdocs inducted into Bouchet Society , The Hub at JHU	May 2026
Q&A with Provost's Postdoctoral Fellow Michael West Jr. , Johns Hopkins MechE	December 2024
Johns Hopkins welcomes new cohort of Provost's Postdoctoral Fellows , The Hub at Johns Hopkins	September 2024
MechE researchers pioneer human motor control theory through global collaboration at TU Munich , MIT MechE	July 2024
A. Michael West: Advancing human-robot interactions in health care , MIT News	September 2023

TECHNICAL SKILLS

- Hardware: rapid prototyping using 3D printers, laser cutters, & machine shop tools (mill & lathe).
- Software: CAD (Solidworks, Creo Parametric), Programming (C, C++, Matlab, Python, Arduino, Machine Learning), MuJoCo, Drake, Visual Studio, Microsoft Office.

REFERENCES

Professor Jeremy Brown Department of Mechanical Engineering Johns Hopkins University 119 Hackerman Hall 3101 Wyman Park Dr, Baltimore, MD 21218 410-516-6609 jdelainebrown@jhu.edu	Professor Neville Hogan Room 3-146 Department of Mechanical Engineering Massachusetts Institute of Technology 77 Massachusetts Avenue Cambridge, Massachusetts 02139 617-253-2277 neville@mit.edu
Professor Meghan Huber Room 207C Department of Mechanical and Industrial Engineering University of Massachusetts, Amherst 160 Governors Drive Amherst, MA 01003 413-545-7656 mehuber@umass.edu	Professor Asegun Henry Room 3-164 Department of Mechanical Engineering Massachusetts Institute of Technology 77 Massachusetts Avenue Cambridge, Massachusetts 02139 617-258-5808 ase@mit.edu